

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and optimized **deep learning models** for neural data analysis, achieving a **15% improvement** in decoding accuracy.
- Collaborated with research scientists to advance **generative modeling** initiatives, contributing to representation learning frameworks.
- Engineered and maintained **machine learning pipelines** for high-performance training on **distributed GPU systems**, ensuring 99.8% uptime.
- Supported the deployment and adaptation of **AI models** across diverse computing environments, addressing evolving business needs and infrastructure updates.
- Authored technical documentation and contributed to experimentation and **hypothesis testing** for research activities, leading to a **20% reduction** in model convergence time.
- Implemented **software engineering best practices**, including automation and code quality checks, enhancing project reproducibility by **30%**.

SKILLS

Machine Learning – Deep Learning, Generative AI, Representation Learning, Supervised Learning, Self-supervised Learning, Neural Data Analysis, Model Optimization, Hypothesis Testing, Experiment Design, Model Evaluation, Large Language Models, Transformers, Diffusers, Accelerator.

Programming & Frameworks – Python, PyTorch, Hugging Face, Scikit-learn, NumPy, Pandas, R.

Data Engineering & Pipelines – ETL, Data Warehousing, Data Pipelines, SQL, PostgreSQL, MySQL, Oracle.

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, GPU Systems.

Development Tools & Methodologies – Git, Jenkins, CI/CD, Agile, Scrum, Jira, Confluence, REST APIs, Linux Shell Scripting.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India

01/2021

PROJECTS

Neural Activity Decoding Engine

- Developed a **PyTorch**-based deep learning model to decode neural signals from simulated brain-computer interface data, achieving **88% classification accuracy**.
- Engineered data preprocessing pipelines using **Python** and **NumPy** to handle large, unstructured neural datasets, reducing data loading times by **25%**.

- Implemented experiment tracking and model versioning using **MLflow** and **Git**, ensuring reproducibility for research iterations.

Generative AI for Synthetic Data Synthesis

- Built and trained a **generative adversarial network (GAN)** using **PyTorch** to synthesize realistic synthetic datasets for augmenting limited real-world data.
- Optimized model architecture and training parameters, resulting in a **10% improvement** in data fidelity metrics compared to baseline models.
- Integrated the model into a **Python**-based workflow, enabling automated generation of synthetic data batches for downstream ML tasks.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)